Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

Problem 1 (2 pts)**.** (a) For which n is the complete graph K_n Eulerian? When does K_n contain an Eulerian trail?

(b) For which s, t is the complete bipartite graph $K_{s,t}$ Eulerian? When does $K_{s,t}$ contain an Eulerian trail?

Solution. (a) The clique K_n is connected, and the degree of any vertex is $n-1$. Thus K_n is Eulerian if and only if $n-1$ is even, i.e., if and only if n is odd. On the other hand, for K_n to have exactly two vertices of even degree we need $n-1=1$. Thus K_n has an Eulerian trail if and only if $n=2$.

(b) The complete bipartite graph $K_{s,t}$ is connected for any positive integers s, t. The degrees of the vertices are s and t, thus $K_{s,t}$ is Eulerian if and only if s and t are both even positive integers. Now, for $K_{s,t}$ to have exactly two vertices of odd degree, we need either $s=t=1$, or $s=2$ and t odd (here, we can interchange the roles of s and t without loss of generality because $K_{s,t} \cong K_{t,s}$. Thus the only complete bipartite graphs that have an Eulerian trail are $K_{1,1}$ (which is just P_2) and $K_{2,t}$ where t is a positive odd integer. □

Problem 2 (2 pts)**.** Let G be a connected regular graph that is not Eulerian. Show that, if $\overline{G}$ is connected, then $\overline{G}$ is Eulerian. Deduce that the complement of the Petersen graph is Eulerian.

Solution. Let G have n vertices and be regular of constant degree d. Since G is connected, the only way it violates the conditions of being Eulerian (Theorem 6.1) is if d is odd. By the handshaking lemma,

$$dn = \sum_{v \in V(G)} \deg_G v = 2|E(G)|$$

is even, which forces n to be even. Now, the graph $\overline{G}$ is $(n-1-d)$-regular. Since n is even and d is odd, $n-1-d$ is even; since $\overline{G}$ is assumed connected, it satisfies the conditions of Theorem 6.1, giving G is Eulerian. □

Problem 3 (2pts)**.** Use Eulerian circuits to show that if G is a connected even-regular graph (i.e. all vertices have even degree), then G has no bridges.

Solution. Note that if G has no edges, then certainly G has no bridges, so we may suppose G has at least one edge.

Fix any edge $e \in E(G)$; we must show that $G-e$ is also connected. Since G is connected and all vertices have even degree, we know that G contains an Eulerian circuit, say $C = (v_0, v_1, \ldots, v_n = v_0)$ (we don't need to specify the edges because G is an ordinary graph). Since the edge e is traversed exactly once in this circuit, we may suppose, without loss of generality, that $e = v_{n-1}v_n$. Now, consider the walk $(v_0, v_1, \ldots, v_{n-1})$. Since $v_n = v_0$, this walk sees all vertices of $G-e$, thus we conclude that $G-e$ is connected. □

Problem 4 (2 + 2 pts)**.** Let G and H be any graphs. Recall that the *Cartesian product* of G and H is the graph $G \times H$ which has vertex set $V(G) \times V(H)$ and $\{(u_1, v_1), (u_2, v_2)\} \in E(G \times H)$ if and only if either $u_1 = u_2$ and $v_1v_2 \in E(H)$ or $u_1u_2 \in E(G)$ and $v_1 = v_2$.

(a) Prove that $G \times H$ is connected if and only if both G and H are connected.

(b) Prove that $G \times H$ is Eulerian if and only if both G and H are connected and also:

- Both G and H are even-regular (i.e. $\deg_G u$ and $\deg_H v$ are always even), or
- Both G and H are odd-regular (i.e. $\deg_G u$ and $\deg_H v$ are always odd).

Solution. (a) ($\Rightarrow$) We show the contrapositive statement: if G or H is disconnected, then $G \times H$ is disconnected. Since $G \times H \cong H \times G$ (as witnessed by the isomorphism $(u, v) \mapsto (v, u)$) we can assume without loss of generality that G is disconnected. Thus we have a partition $V(G) = A \sqcup B$ where A and B are non empty subsets such that no edge in G goes between A and B. But then for the Cartesian product $G \times H$ we have

$$V(G \times H) = V(G) \times V(H) = (A \times V(H)) \sqcup (B \times V(H)).$$

We claim that there is no edge in $G \times H$ that crosses between $A \times V(H)$ and $B \times V(H)$, which will imply that $G \times H$ is disconnected because these pieces are non-empty. Consider arbitrary elements $(a, v) \in A \times V(H)$ and $(b, w) \in B \times V(H)$, and let us show that $\{(a, v), (b, w)\}$ is *not* an edge in $G \times H$. Indeed, since A and B are disjoint we have $a \neq b$, thus the only way that $\{(a, v), (b, w)\} \in E(G \times H)$ □